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CAMBRIDGE
INTERNATIONAL EXAMINATIONS

November 2003

INTERNATIONAL GCSE

MARK SCHEME

MAXIMUM MARK: 60

SYLLABUS/COMPONENT: 0625/05

PHYSICS
Practical

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1	**(b)(c)**	Table A, 6 temps, decreasing	
		Table B, 6 temps, decreasing	1
		Temp unit	1
		Time unit	1
		Evidence of temp to better than 1°C	1
		Consistently better than 1°C	1
	(d)	Graph:	
		Time axis suitable (no '3' scales allowed)	1
		Time axis labeled	1
		Check plots at 210 s and 240 s	1
			1
		lines judgement (best fit curves)	1
		lines thickness	1
		Both lines correctly labeled	1
	(e)	Conclusion:	
		Correct statement in relation to candidate's lines	1
		Explained with correct reference to gradients (if previous mark scored)	1
			TOTAL 15
2	**(b)**	x = 20.0 (cm)	1
	(c)	y value less than 25 cm	1
		y value to nearest mm	1
	(d)	d = 25 (cm) (allow e.c.f.)	1
	(e)	t value correct arith	1
	(f)	x = 30 (cm)	1
		y value in range 30.0 – 37.5 (cm)	1
		d = 37.5 (cm) (allow e.c.f.)	1
		all x, y, d consistently in mm, cm or m (unit stated at least once)	1
		x, y d units stated every time	1
		t value correct arith	1
		t values within 0.5 cm of each other	1
	(g)	average t; correct method	1
		final answer to 2/3 sf	1
		with correct unit	1
			TOTAL 15

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3		Trace	
		Neat thin lines	1
		Lines complete	1
		A and B correct positions	1
		New B correct	1
		i = r (by eye)	1
		CD at least 5 cm	1
		Second CD at least 5 cm	1
		Straight lines extended to X	1
		XA drawn and Y labeled	1
	(j)	AY correct to 2 mm	1
		YX correct to 2 mm	1
		AY and YX same to within 10 mm	1
	(k)	Thickness of mirror OR thickness of pins OR thickness of lines	1
	(l)	Precaution (pin separation, view bases, vertical pins)	1
		Reason	1
			TOTAL 15
4.	**(b)–(g)**	x in m, cm or mm	1
		V in V	1
		k in V/m, V/cm or V/mm	1
		correct x values (0.200, 0.400, 0.800 m)	1
		all x to nearest mm	1
		x consistent sf	1
		evidence of V to better than 0.5 V	1
		all V to better than 0.5 V	1
		3 k values	1
		Check second k value, correct	1
		all k to 2 sf OR all k to 3 sf	1
		all k same to within 10%	1
	(h)	(voltage increases with length)	1
		OR voltage proportional to length	2
		k = constant OR figures correctly quoted	1
			TOTAL 15